Lesson plan · for the teacher

Circuits and microcontrollers: unit check

IoT · Level 2 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Current, LEDs and resistors
- Logic: switches that decide
- Bits and bytes
- Microcontrollers: sense, decide, act

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it in class tomorrow

Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A torch with new batteries won't light. What should you check first?**

- ◯ The loop: wires, contacts and battery direction
- ◯ Whether a bigger battery would push more current
- ◯ Whether the bulb needs a resistor to protect it

2 **Match each wiring or device to its gate.**

Two switches in series	• • • •	XOR
Two switches in parallel	• • • •	NOT
A fridge-door light switch	• • • •	OR
A stair light with a switch at each end	• • • •	AND

3 **Choose the resistor for a red LED on two AA batteries.**

3 V from the batteries minus 2 V for the LED leaves _____ V. For 10 mA, the resistor should be _____ Ω.

Word bank: 1 · 100 · 3 · 300

4 **A garden pump runs when "soil dry AND NOT raining". The soil is dry and it is raining, so the pump runs.**

◯ True ◯ False

5 **Which statements about a microcontroller are true?**

- ☐ Setup runs once, at power-on
- ☐ The loop runs again and again
- ☐ An analog pin turns a voltage into a number
- ☐ A pin can power a pump directly
- ☐ It needs the internet to work